

■■ RealEyes AI — Comprehensive Technical Report

Project Title: RealEyes AI — Deepfake Detection using Vision Transformers, Temporal Video & Audio Analytics

Repository: <https://github.com/Prasad-creates/realeyes> (<https://github.com/Prasad-creates/realeyes>)

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Document Status: Final Technical Specification & Architecture Report

■ 1. Executive Summary

RealEyes AI is an enterprise-grade, multi-modal synthetic media detection and identity verification platform engineered to detect AI-generated deepfake images, manipulated video streams, and synthetic voice clones in real-time.

As digital identity verification (KYC/AML) processes increasingly rely on remote media submission, synthetic media powered by diffusion models (Midjourney, DALL-E 3, Flux), GANs, and neural vocoders (ElevenLabs, Bark) poses a critical security threat. RealEyes AI mitigates this threat by combining **Vision Transformer (ViT) deep learning**, **Error Level Analysis (ELA) heuristics**, **temporal video frame sampling**, and **Wav2Vec2 audio spectral analysis** into a unified, explainable forensic defense suite.

The platform provides three primary interfaces:

- 1. FastAPI AI Engine:** Python 3.10 microservice running PyTorch & HuggingFace Transformer models.
- 2. React 19 Web Portal:** High-performance dashboard with drag-and-drop media upload, live analysis visualizer, reverse image search, and automated DOCX forensic report generation.
- 3. Chrome Extension (Manifest V3):** Browser-injected scanner with on-page element inspection shields, live canvas region selection, and offscreen audio capture.

■■ 2. System Architecture

```
graph TD
    subgraph Client_Layer
        A[Chrome Extension Manifest V3] -->|API Requests| B[React Web Portal Vite]
        B -->|HTTP / FormData| C[Express API Middleware Port 5001]
    end

    subgraph Middleware_and_Business_Logic
        C -->|SHA-256 Hashing| D[(In-Memory Cache & MongoDB)]
        C -->|HTTP Proxy| E[FastAPI AI Server Port 8000]
    end

    subgraph AI_Inference_Engine
        E --> F[Vision Transformer Model ViT]
        E --> G[ELA & Frequency Domain Heuristics]
        E --> H[OpenCV Temporal Video Keyframe Sampler]
        E --> I[Wav2Vec2 Audio Neural Vocoder Detector]
    end

    subgraph External_Services
        B -->|DOCX Generation| J[Report Download Engine]
        B -->|Reverse Search| K[Google Lens / TinEye / Yandex / Bing]
    end
```

■ 3. Core Modules & Technical Implementation

3.1. Vision Transformer (ViT) Image Detector

- **Model Architecture:** **dima806/deepfakevsrealimagedetection** (Vision Transformer patched at 16x16 resolution).
- **Functionality:** Evaluates global self-attention maps to identify micro-boundary anomalies, GAN grid artifacts, and dermal texture inconsistencies invisible to the human eye.
- **Error Level Analysis (ELA):** Re-compresses images at known JPEG quality scales (90%) and computes differential error maps to detect localized re-saving or pixel manipulation boundaries.

3.2. Temporal Video Frame Analyzer

- **Sampling Mechanism:** Extracts keyframes using OpenCV at uniform temporal intervals across the video stream.
- **Timeline Scrubber:** Generates 10 discrete keyframe segment confidence scores, identifying exact timestamps where synthetic face-swapping or generative diffusion was introduced.

3.3. Audio Voice Clone Detector

- **Model Architecture:** Wav2Vec2 / Audio Spectrogram Transformer.
- **Analytical Metrics:** Evaluates acoustic spectral periodicity, phase alignment, and phase-lock vocoder signatures to identify synthetic voice synthesis from models such as ElevenLabs or Bark.

3.4. Web Application Portal (Vite + React 19)

- **Styling & UI:** Custom dark glassmorphism theme built with Vanilla CSS and Tailwind CSS, featuring subtle glows (**glow-red**, **glow-green**), reactive status rings, and smooth micro-animations.

- **DOCX Forensic Report Exporter:** Uses **docx** and **file-saver** to client-side compile cryptographic SHA-256 hashes, timestamped verdicts, and model confidence scores into downloadable formal forensic reports.
- **Reverse Image Search Integration:** Built-in triggers for Google Lens, TinEye, Yandex Images, and Bing Visual Search using temporary cloud blob uploads.

3.5. Chrome Browser Extension (Manifest V3)

- **On-Page Image Shields:** Content script (**imageScanner.ts**) inspects DOM **** elements and overlays interactive shield indicators. Clicking a shield opens a draggable analysis modal directly over the webpage.
- **Region Selector:** Live canvas overlay (**regionSelector.ts**) allows users to crop and scan any screen section in real-time.
- **Offscreen Audio Recording:** Utilizes Chrome Extension Offscreen API (**offscreen.html / offscreen.ts**) to intercept tab audio streams without interrupting user playback.

■ 4. API Reference Table

Base URL: **http://localhost:8000** (FastAPI AI) | **http://localhost:5001** (Express Gateway)

Endpoint	Method	Input Payload	Response Output	Description
/analyze/image	POST	multipart/form-data (file)	JSON (label, confidence, scores, details)	Performs ViT + ELA deepfake analysis on uploaded image file.
/analyze/image-url	POST	Form (url)	JSON (label, confidence, scores, details)	Fetches remote image URL and executes detection.
/analyze/image-base64	POST	JSON (image, mime)	JSON (label, confidence, scores, details)	Decodes base64 payload for Chrome extension region captures.
/analyze/video	POST	multipart/form-data (file)	JSON (isFake, score, timelineSegments)	Samples video keyframes and calculates frame-by-frame authenticity.
/analyze/audio	POST	multipart/form-data (file)	JSON (label, confidence, scores)	Analyzes audio waveforms for synthetic voice clone signatures.
/api/tools/feedback	POST	JSON (hash, userVerdict)	JSON (success: true)	Submits human verdict feedback for RLHF pipeline training.

■ 5. Key Improvements Completed

1. **Unrestricted Open Access:** Removed authentication restrictions (Google OAuth / password walls) so all features and detection tools are immediately accessible to users.

2. **Pre-Cached AI Weights:** Downloaded and verified HuggingFace Vision Transformer model weights (~350 MB) to ensure sub-second inference speeds.
3. **Timeout & Fallback Protection:** Implemented **AbortController** request timeouts (8s backend, 10s frontend) with deterministic fallback heuristics to guarantee the web interface never hangs.
4. **Git & GitHub Integration:** Configured clean versioning and pushed all source code to <https://github.com/Prasad-creates/realeyes> (<https://github.com/Prasad-creates/realeyes>).

■ 6. Recommendations & Future Enhancements

1. **GPU Acceleration (CUDA):** Deploy the FastAPI backend on an NVIDIA GPU instance (e.g. AWS g4dn or RunPod) to enable 60 FPS video stream inference.
2. **Explainable AI (Grad-CAM):** Add visual heatmaps highlighting suspicious facial bounding boxes directly on the React image preview.
3. **Continuous Audio Stream Detection:** Expand the Chrome Extension to continuously monitor WebRTC video call streams (e.g. Google Meet, Zoom web) for real-time deepfake alerts.